CSE 403

Artificial Intelligence & Expert Systems

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22101040 | Section: A

Semester: Spring 25

Contents

Chapter 1

Knowledge Representation and Reasoning

Rules of Inference

Modus Ponens	Modus Tollens	Disjunctive Syllogism	Hypothetical Syllogism
$\dfrac{P \to Q \quad P}{\therefore Q}$	$\dfrac{P \to Q \quad \neg Q}{\therefore \neg P}$	$\dfrac{P \lor Q \quad \neg P}{\therefore Q}$	$\dfrac{P \to Q \quad Q \to R}{\therefore P \to R}$

Examples

Question-1: Represent the following knowledge statements using propositional logic. Apply rules of inference to conclude **"The laptop is working"**.

1. Either the laptop is working or I will study at the library.
2. If I study at the library, then I will not attend the online seminar.
3. I attended the online seminar.

Solution:

Let propositions are,

P : The laptop is working.

Q : I will study at the library.

R : I will attend the online seminar.

Given Statements,

$P \lor Q \quad \ldots(1)$

$Q \to \neg R \quad \ldots(2)$

$R \quad \ldots(3)$

From statement (3) and (2), by Modus Tollens, we get,

$$\begin{array}{l} Q \to \neg R \\ R \\ \hline \therefore \neg Q \ldots\ldots(4) \end{array}$$

From statement (1) and (4), by Disjunctive Syllogism, we get,

$$\begin{array}{l} P \lor Q \\ \neg Q \\ \hline \therefore P \end{array}$$

Question-2: Represent the following knowledge statements using propositional logic. Apply rules of inference to conclude **"Applications can be processed"**.

1. If Admission office is open, Staff are present
2. If Staff are present, Applications can be processed
3. If emergency drill is active, the building is evacuated
4. The emergency drill is not active
5. The admission office is open or the building is evacuated.

Solution:

Let propositions are,

P : Admission office is open.

Q : Staff are present.

R : Applications can be processed.

S : Emergency drill is active.

T : Building is evacuated.

Given Statements,

$P \to Q \quad \ldots(1)$

$Q \to R \quad \ldots(2)$

$S \to T \quad \ldots(3)$

$\neg S \quad \ldots(4)$

$P \lor T \quad \ldots(5)$

From statement (4) and (3), by Modus Tollens, we get,

$S \to T$

$\neg S$

$\therefore \neg T \ldots\ldots(6)$

From statement (5) and (6), by Disjunctive Syllogism, we get,

$P \lor T$

$\neg T$

$\therefore P$

From statement (1) and (7), by Modus Ponens, we get,

$P \to Q$

P

$\therefore Q$

From statement (2) and (8), by Modus Ponens, we get,

$Q \to R$

Q

$\therefore R$

Thus, we conclude that **"Applications can be processed"**.

Chapter 2

Predicate Logic

Question: Represent the following knowledge using First Order Predicate Logic (FOPL): David, William, and Kim are members of the CSE Sports Club. Every member of the Sports Club who is not a footballer is a cricketer. Cricketers do not like rain, and anyone who does not like rain is not a footballer. David dislikes whatever Kim likes, and likes whatever Kim dislikes. William likes rain and snow.

Predicates:
Member(x): x is a member of the CSE Sports Club.
Footballer(x): x is a footballer.
Cricketer(x): x is a cricketer.
Likes(x, y): x likes y.

FOPL Representation:

1. David, William, and Kim are members of the CSE Sports Club.
   Member(David) $\wedge$ Member(William) $\wedge$ Member(Kim)
2. Every member of the Sports Club who is not a footballer is a cricketer.
   $\forall x$ (Member(x) $\wedge$ $\neg$Footballer(x) $\rightarrow$ Cricketer(x))
3. Cricketers do not like rain.
   $\forall x$ (Cricketer(x) $\rightarrow$ $\neg$Likes($x, Rain$))
4. Anyone who does not like rain is not a footballer.
   $\forall x$ ($\neg$Likes($x, Rain$) $\rightarrow$ $\neg$Footballer(x))
5. David dislikes whatever Kim likes.
   $\forall y$ (Likes(Kim, y) $\rightarrow$ $\neg$Likes($David, y$))
6. David likes whatever Kim dislikes.
   $\forall y$ ($\neg$Likes(Kim, y) $\rightarrow$ Likes($David, y$))
7. William likes rain and snow.
   Likes(William, Rain) $\wedge$ Likes(William, Snow)

Chapter 3

Genetic Algorithm

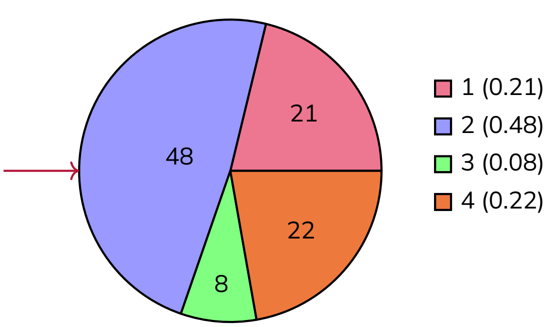
Question: Simulate one step of Genetic Algorithm (selection, crossover and mutation) for the following function. Represent the population, x using 6 bits. Consider the population size as $F(x) = 2x^2 + 3; 0 \leq x \leq 63.$

Solution:

Initialization,
Number of chromosomes = 4
Number of bits/gene in each chromosome = 6
Let, $C1$: 100011 $C2$: 110101 $C3$: 010110 $C4$: 100100

Selection:

String no	Initial Population	x Value	Fitness Value $F(x) = 2x^2 + 3$	Probability$_i$	Expected Count	Actual Count
1	100011	35	2453	0.21	0.84	1
2	110101	53	5621	0.48	1.93	2
3	010110	22	971	0.08	0.33	0
4	100100	36	2595	0.22	0.89	1
Sum			11640	1.00	4.00	4
Average			2910	0.25	1.00	1
Max			5621	0.48	1.93	2



Crossover:

String no.	Mating Pool	Crossover Point	Offspring	x value	Fiteness $(2x^2+3)$
1	100011	3	100101	37	2741
2	110101	3	110011	51	5205
2	110101	3	110100	52	5411
4	100100	3	100101	37	2741
Sum					16098
Average					4024.5
Max					5411

Mutation:

String no.	Offspring after crossover	Offspring after mutation	x value	Fitness $(2x^2+3)$
1	100101	110101	53	5621
2	110011	110011	51	5205
2	110100	110100	52	5411
4	100101	110101	53	5621
Sum				21858
Average				5464.5
Max				5621

Chapter 4

Artificial Neural Network

Formulas

Activation functions:

Step function:	$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$
Sigmoid function:	$f(x) = \frac{1}{1+e^{-x}}$
Tanh function:	$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$
ReLU function:	$f(x) = \max(0, x)$

Output	: $y_p = \sigma\left(\sum w_i x_i - \theta\right)$
Error	: $e = y_t - y_p$
Error gradient (output layer)	: $\delta = y \cdot [1-y] \cdot e$
Error gradient (hidden layer)	: $\delta_i = y_i \cdot [1-y_i] \cdot \sum \delta_j w_{ij}$
Weight update	: $w'_{ij} = w_{ij} + \alpha \delta_j y_i$

Question: For the following perceptron, the feature vector is X = [1 0 1] and the desired output is $Y = 1$. Consider the threshold value $\theta = 0.3$, learning rate $\alpha = 0.1$, initial weights $W_1 = 0.3$, $W_2 = 0.5$, and $W_3 = 0.6$. i) Measure the predicted output and ii) Calculate the updated weights (Wi, Wz and W3) after one iteration.

Solution:

For $P = 1$,

$$Y' = \text{Step}([(W1 * X1 + W2 * X2 + W3 * X3) - \theta])$$

$$= \text{Step}(0.3 \times 1 + 0.5 \times 0 + 0.6 \times 1) - 0.3 = 0.6$$

$$= \text{Step}(0.6) = 1$$

$\therefore$ Error $E = Y - Y' = 1 - 1 = 0$

Since there is no error, no weight update is needed.

Weight update:

$$w'_1 \quad = w_1 + \alpha x_1 E = 0.3 + 0.1 \times 1 \times 0 = 0.3$$

$$w'_2 \quad = w_2 + \alpha x_2 E = 0.5 + 0.1 \times 0 \times 0 = 0.5$$

$$w'_3 \quad = w_3 + \alpha x_3 E = 0.6 + 0.1 \times 1 \times 0 = 0.6$$

For the following back propagation neural network shown in Figure 2, the feature vector is X = [1 1] and the desired output is $Y = 0$. Consider the threshold value $\theta_1 = 0.5$, $\theta_2 = -1.5$, $\theta_3 = 1.5$, learning rate $\alpha = 0.5$, initial weights $W_1 = 1$, $W_2 = -0.5$, $W_3 = 1$, $W_4 = 1$, $W_5 = 1$, and $W_6 = 1$.

1. Measure the predicted outputs of the hidden layer (neuron h_1 and h_2).
2. Measure the predicted output of the output layer (neuron o_1).
3. Calculate the updated weights of output layer (W_5 and W_6) after one iteration.

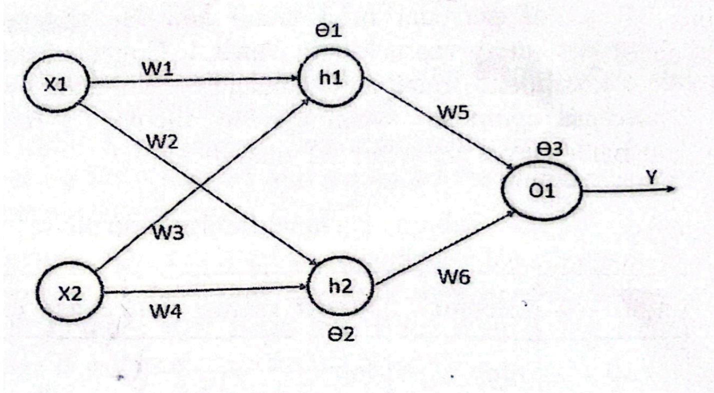


Figure 2. Back Propagation Neural Network

Solution:

For hidden layer:

$$Y_{h_1} = \sigma\left[(W_1X_1 + W_3X_2) - \theta_1\right] = \sigma\left[(1 \times 1 + 1 \times 1) - 0.5\right] = \sigma(1.5) = \frac{1}{1 + e^{-1.5}} = 0.8176$$

$$Y_{h_2} = \sigma\left[(W_2X_1 + W_4X_2) - \theta_2\right] = \sigma\left[(-0.5 \times 1 + 1 \times 1) - (-1.5)\right] = \sigma(2) = \frac{1}{1 + e^{-2}} = 0.8807$$

For output layer:

$$Y_{o1} = \sigma\left[(W_5Y_{h_1} + W_6Y_{h_2}) - \theta_3\right] = \sigma\left[(1 \times 0.8176 + 1 \times 0.8807) - 1.5\right] = \sigma(0.1983) = 0.5494$$

Now calculating error at output level,

$$E_{o1} = Y - Y_{01} = 0 - 0.5494 = -0.5494$$

Now calculating error gradient,

$$\delta_{o1} = Y_{o1} \cdot [1 - Y_{o1}] \cdot E_{o1} = 0.5494 \cdot [1 - 0.5494] \cdot (-0.5494) = -0.136$$

Now updating weight between output and hidden layer,

$$W_5' = W_5 + \alpha Y_{o1}\delta_{o1} = 1 + 0.5 \times 0.5494 \times -0.136 = 0.96$$

$$W_6' = W_6 + \alpha Y_{o1}\delta_{o1} = 1 + 0.5 \times 0.5494 \times -0.136 = 0.96$$

Error gradients at hidden layer,

$$\delta_{h1} = Y_{h1} \cdot [1 - Y_{h1}] \cdot [W_5\delta_{o1}] = 0.8176 \cdot [1 - 0.8176] \cdot [1 \times -0.136] = -0.02$$

$$\delta_{h2} = Y_{h2} \cdot [1 - Y_{h2}] \cdot [W_6\delta_{o1}] = 0.8807 \cdot [1 - 0.8807] \cdot [1 \times -0.136] = -0.01$$

Now updating weight between hidden and input layer,

$W_1' = W_1 + \alpha X_1 \delta_{h1} = 1 + 0.5 \times 1 \times -0.02 = 0.99$

$W_2' = W_2 + \alpha X_2 \delta_{h2} = -0.5 + 0.5 \times 1 \times -0.02 = -0.51$

$W_3' = W_3 + \alpha X_1 \delta_{h1} = 1 + 0.5 \times 1 \times -0.01 = 0.995$

$W_4' = W_4 + \alpha X_2 \delta_{h2} = 1 + 0.5 \times 1 \times -0.01 = 0.995$

Chapter 5

Uncertainty

Inference using Joint Distributions

For the following joint distribution, find the conditional probability of $P(cavity|\neg toothache)$. Indicate query variable, evidence variable and hidden variable. Where the values of joint distribution are $A = 0.324$, $B = 1 - A$, $C = 0.576$ and $D = 0.144$.

	¬ toothache	
	catch	¬ catch
cavity	0.324	0.676
¬ cavity	0.576	0.144

Solution:

$$P(cavity|\neg toothache) = \frac{P(cavity \cap \neg toothache)}{P(\neg toothache)} = \frac{0.324 + 0.676}{0.324 + 0.676 + 0.576 + 0.144} = \frac{1}{1.72}$$

Naive Bayes